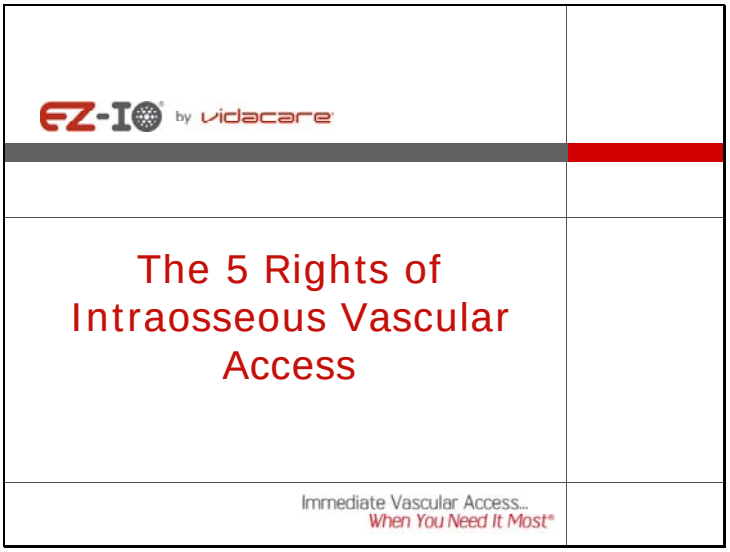
Slide 1

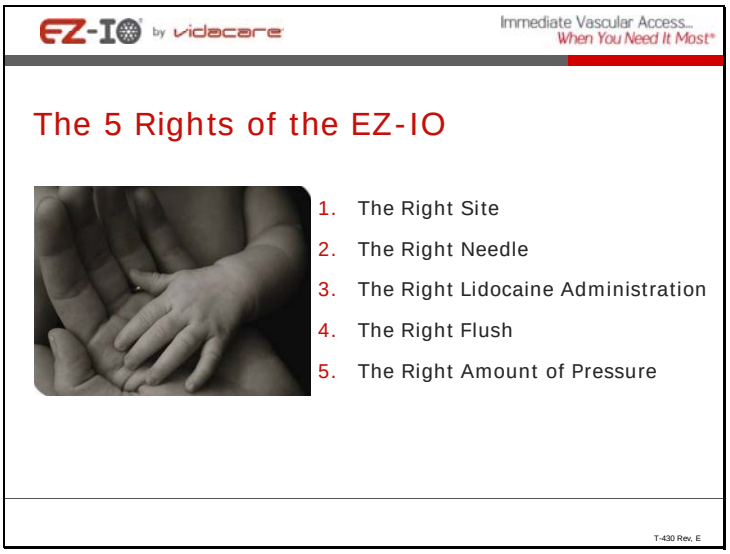


Thank you for taking the time to participate in Vidacare's training program on the EZ-IO vascular access system. This training program is designed to provide the practitioner with the key skills and knowledge to be competent inserting the EZ-IO Needle Set, while understanding the concepts of:

- Which patients are candidates for an EZ-IO
- What are the IO contraindications
- Basic bone anatomy
- Basic physiology such as how fluid gets from intraosseous space to the venous system
- Identification of insertion landmarks
- Care and maintenance of the EZ-IO
- Removal of the device and after care

This program is designed to be used in concert with hands-on practice with the EZ-IO device, various needle sets, and an array of simulation insertion aids such as synthetic bones of varying density. If at any time you have a question that needs clarification, please feel free to contact the Clinical Manager in your area to get a clear, concise and thorough answer. You can find the contact information for your Clinical Manager at www.vidacare.com or by calling 1-866- 479-8500.

Slide 2



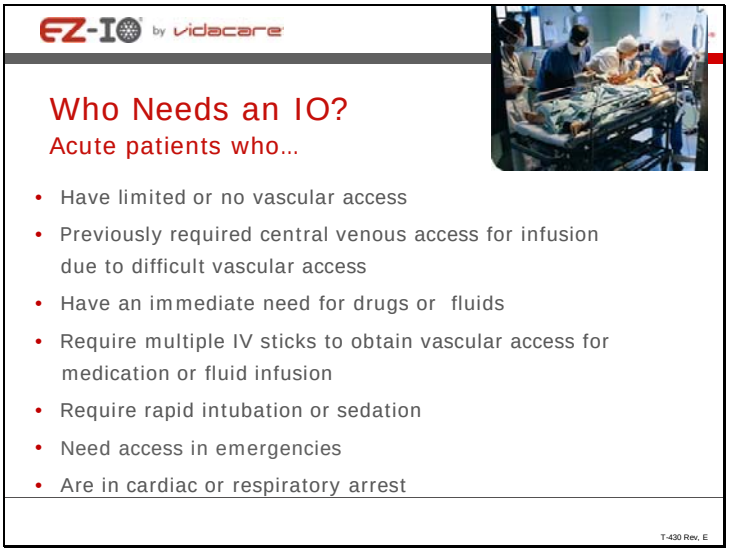
Vidacare employs the concept of the 5 Rights, as this allows the practitioner to easily recall the 5 basic principles of the EZ-IO system.

<u>The 5 Rights are:</u>

- The Right Site
- The Right Needle
- The Right Lidocaine Administration
- The Right Flush
- The Right Amount of Pressure

A statement of the 5 rights will be accompanied by a more in-depth explanation of each concept that ensures competent and safe use of the EZ-IO system.

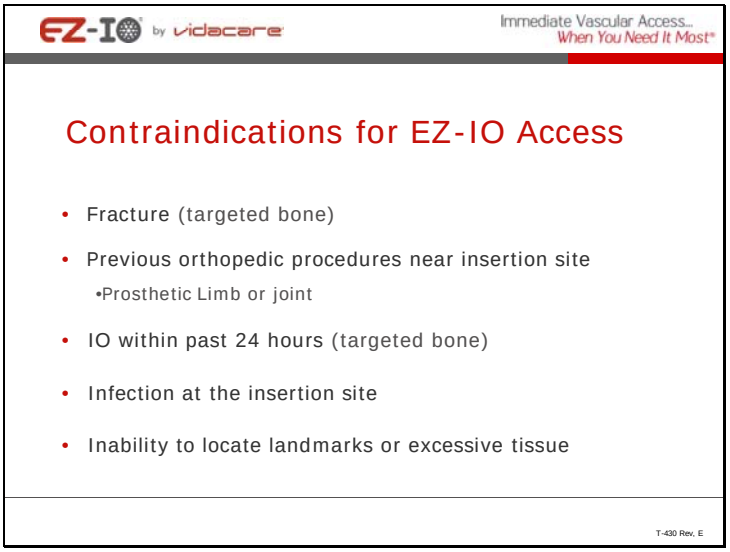
Slide 3



Before we explore how to use the EZ IO system, we need to decide which patients are ideal candidates for an EZ-IO. Traditionally, intraosseous access was reserved as a last ditch effort in children in cardiac or respiratory arrest or children in extremis. Due to tremendous advances in intraosseous technology, clinicians today are able to rapidly gain access to the intraosseous space in both adults and children alike. Additionally, the technological advancements achieved by the EZ-IO system allows clinicians to use this device in responsive patients due to the relatively minor pain of insertion.

Ultimately, any patient who exhibits an acute need for fluids or medication is an appropriate candidate for an EZ-IO.

Slide 4



Although there are limitations to IO access, these limitations are usually isolated to the placement of the EZ-IO rather than patient type or size.

The contraindications are as follows:
Fracture (targeted bone): If there is a fracture to the bone in which you want to place the EZ-IO, you must choose an alternate site. Fluid follows the path of least resistance. If an IO is placed in a fractured bone, the fluid would simply extravasate into the surrounding tissue through the fracture site. In the case of the tibia, if there is a fracture present, both the proximal and distal sites are contraindicated as they share the same common pathway inside the bone. However; if you have a fractured femur and in intact tibia, you can still use the tibial sites as it is totally separate from the femoral compartment. Judge the appropriateness of using the lower extremity for EZ-IO the same as you would for any IV or femoral central line. Do not use the EZ-IO if there is a strong possibility of vascular compromise. The proximal humerus is a better anatomical site for EZ-IO access in such cases.

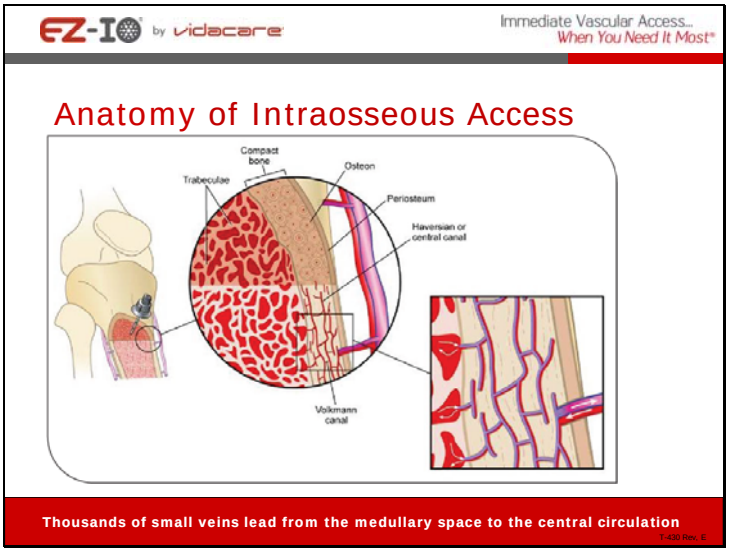
Previous orthopedic procedures near insertion site: The basic rule of thumb is that if you find a surgical scar over a joint; assume there is a titanium appliance within the joint itself. Even though the EZ-IO needle sets are revolutionary, they cannot penetrate titanium. If this is the case, simply select one of the five other FDA approved insertion sites.

IO within past 24 hours (targeted bone): Healing generally takes 24 hours and is defined as the point where another IO can be safely placed at the same anatomical site. By that time, fibrin formation and clotting are sufficient to prevent extravasation through the previous IO hole. Complete healing, to the point where x-ray can no longer detect the hole, usually takes several days.

Infection at the insertion site: Reported infection rates for traditional IO are low. To date, there have been no reported cases of osteomyelitis in more than 300,000 EZ-IO uses. Vidacare does not recommend inserting the EZ-IO through an obvious or documented infection over the insertion site. Acutely burned skin is not a contraindication for IO access. The EZ-IO must be removed within 24 hours.

Inability to locate landmarks or excessive tissue: If you are unable to locate the landmarks for an insertion site, for any reason including excess adipose tissue select an alternate identifiable insertion site.

Slide 5

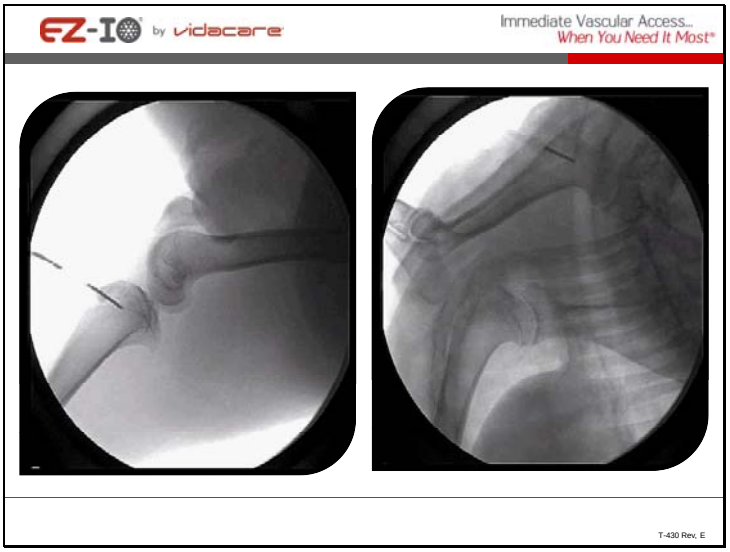


Intraosseous needles are placed in the proximal and distal ends (epiphysis) of long bones such as the tibia and humerus due to the thinner compact bone and abundance of cancellous (spongy) bone found at these sites. Within the micro-vasculature inside the medullary space, lies a hyper-coagulable fibrin mesh as well as red and yellow marrow. Since this is the space where all blood cells derive from, we can follow the pathway from the medullary space to the venous system. Within the epiphysis or medullary space lies a vast system of canals that blood and fluid can travel through to reach the central circulation.

The myth often perpetuated is that it takes a long time for fluids, medications, or blood infused into the intraosseous space to reach the central circulation. To the contrary, any fluid instilled into the intraosseous space gains access to the central circulation within just a few seconds. A study out of UTMB in Galveston by Kramer, et al. measured peak serum concentrations of epinephrine and found that epinephrine infused via the intraosseous humeral site has the identical peak serum concentration as if it were instilled via a subclavian central line.1

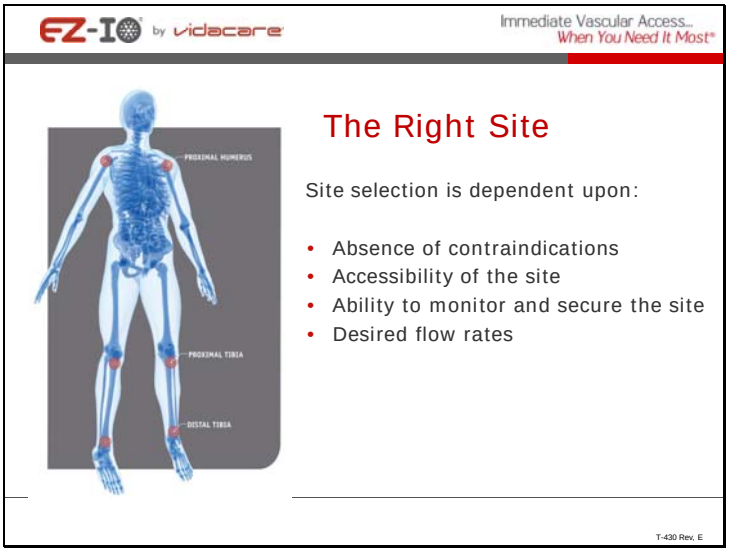
1Kramer GC, Hoskins SL, Espana J, et al. Intraosseous drug delivery during cardiopulmonary resuscitation: relative dose delivery via the sternal and tibial routes. *Acad Emerg Med* 2005;12(5):s67.

Slide 6



These real-time flow rate studies demonstrate how quickly IV contrast infused into the medullary space is disseminated into the central circulation. The video on the left shows dye infused through the proximal tibia. It is rapidly absorbed into the popliteal and femoral vasculature. The video on the right shows how rapidly dye reaches the right atrium when instilled via the proximal humerus, again demonstrating direct correlation between subclavian and proximal humerus for infusion of fluids.

Slide 7

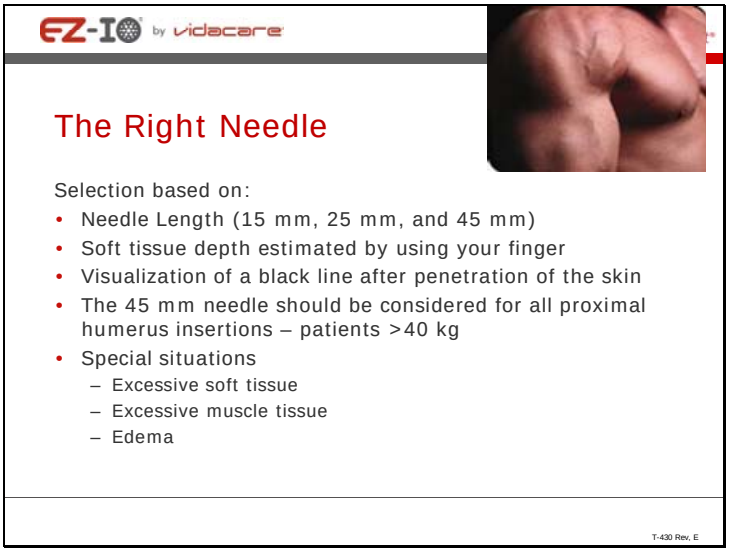


The EZ-IO is approved to be inserted in 6 anatomical sites. These sites are the right and left proximal tibia, the distal tibia, and the proximal humerus.

Site selection is dependent on several factors including:

- **Absence of contraindications** such as fracture of the target bone, local infection, inability to locate the landmarks or scar indicative of prior joint surgery
- **Accessibility of the site**
- **Ability to monitor and secure the site**
- **Desired flow rates-** Traditionally the proximal humerus site delivers higher volumes of fluid per minute. This may be of consideration if fluid volume resuscitation is required.

Slide 8



To ensure success with the EZ-IO system, it is imperative you select the right needle for the patient. To maximize a successful insertion, you need to assess the selected insertion site, as well as the patient's physiologic needs. One misconception is that needles are separated into pediatric, adult and large sizes. In fact, selecting the correct needle set is based on tissue depth overlying the insertion site.

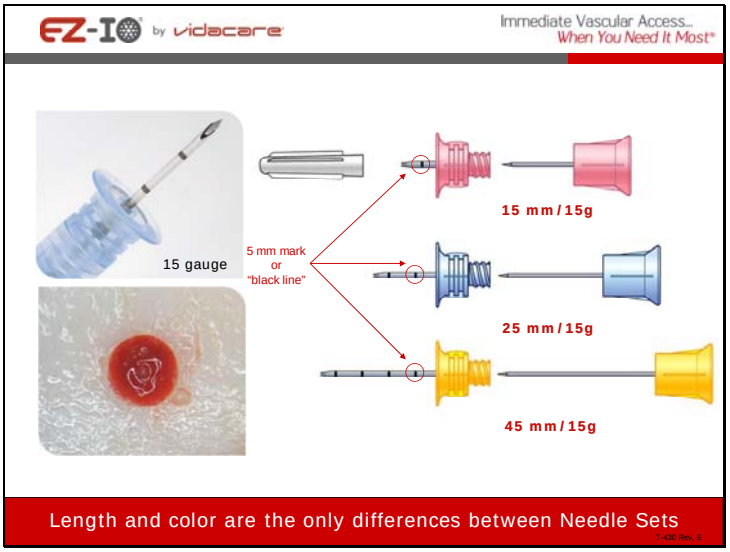
The best way to determine which Needle Set is appropriate for your patient is by assessing tissue depth with your finger or thumb. This is done by palpating the insertion site to get a rough estimate of the depth of tissue. The amount of tissue at the site may range from virtually none (palpate the tissue overlying one of your knuckles) to excessive (palpate the amount of tissue overlying your femur).

The next thing to take into account is any special situations that may require a longer needle such as insertion into the proximal humerus, patients with excessive soft tissue, patients with excessive muscle tissue, or patients with substantial edema. These patients will most likely benefit from use of the 45 mm needle to ensure successful 1st insertion cannulation.

The final aid in helping select the appropriate length needle is the visualization of at least one black line. With the needle attached to the driver, insert the needle through the skin until the tip rests against the bone

at a 90 degree angle. Look at the needle to assure that you can visualize at least one black line above the surface of the skin. If you cannot see a black line, the Needle Set selected is too short to reach the medullary space.

Slide 9



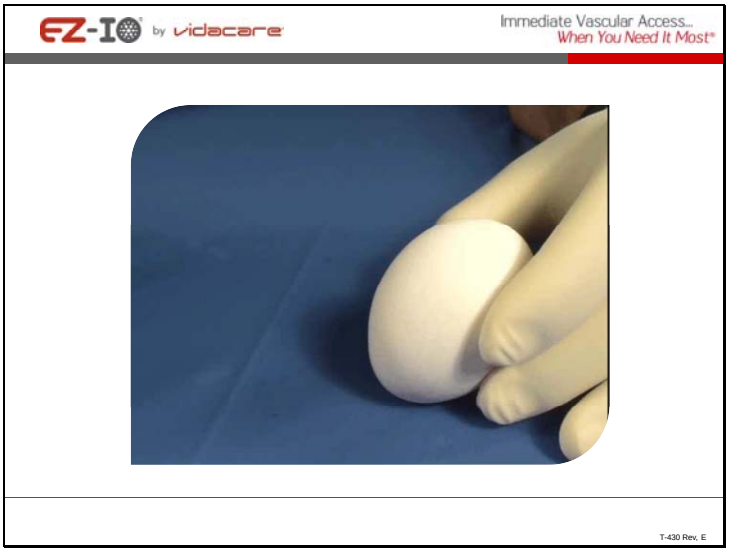
The EZ-IO system has three Needle Sets to choose from.

- The pink needle which is 15 mm in length
- The blue needle which is 25 mm in length
- The yellow needle which is 45 mm in length

All three Needle Sets are 15 gauge.

The EZ-IO needle is a uniquely designed needle tip that cuts into the bone allowing for a relatively painless insertion. This technology creates a hole in the cortex that is the same size as the needle thus minimizing the risk of extravasation and dislodgement. Since the EZ-IO is inserted gently with minimal pressure, the risk of micro-fractures is also greatly reduced if not eliminated.

Slide 10



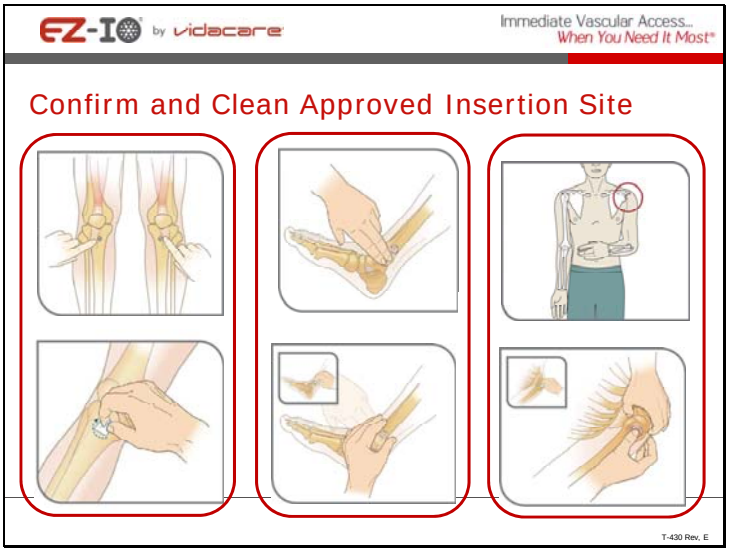
This video portrays the precision, control, and gentleness of the EZIO vascular access system utilizing the thinnest most delicate object – a raw egg.

Slide 11

Any medication that can be safely given through a peripheral vein can safely be infused into the medullary space. There are no dose changes, however; remember to follow each medication administration with a 3-5 ml fluid flush to ensure the entire dose is pushed out of the medullary space and into the central circulation. Use caution with hypertonic saline solutions of longer than 30 minute duration.

Blood draws for laboratory testing are frequently acquired from the medullary space. Since the cavity within the intraosseous space is lined with a viscous fibrin material, it is important to aspirate and discard 2 ml before obtaining a sample for lab analysis. Correlation between medullary samples and venous samples may be done to ensure correct interpretation (label the tubes as IO blood).

Slide 12



This slide demonstrates the 3 approved sites that an IO can be inserted bilaterally creating 6 vascular access points.

The **proximal tibia** insertion site is approximately 2 cm below the patella and approximately 2 cm medial to the tibial tuberosity (depending on patient anatomy).

The **distal tibia** insertion site is located approximately 3 cm proximal to the most prominent aspect of the medial malleolus (depending on patient anatomy). Place one finger directly over the medial malleolus; move approximately 3 cm proximal and palpate the anterior and posterior borders of the tibia to assure that your insertion site is on the flat center aspect of the bone.

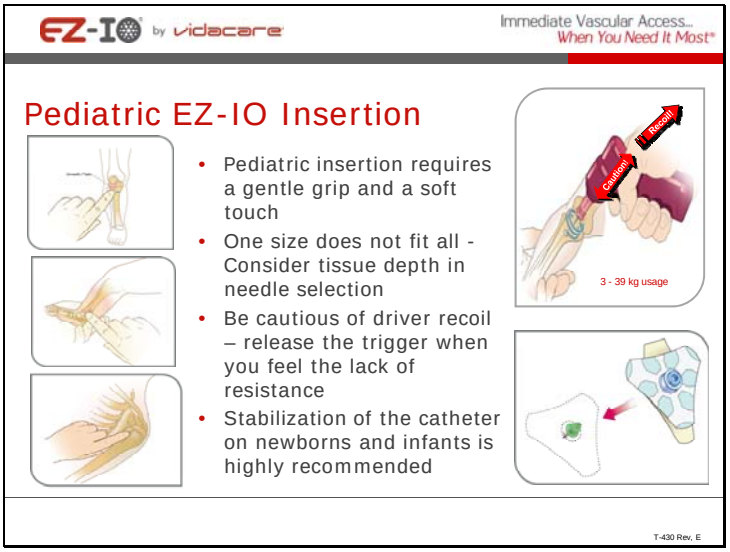
The **proximal humerus insertion** site is located directly on the most prominent aspect of the greater tubercle. *Ensure that the patient's hand is resting on the abdomen and that the elbow is adducted (close to the body).* Slide thumb up the anterior shaft of the humerus until you feel the greater tubercle, this is the surgical neck. Approximately 1 cm (depending on patient anatomy) above the surgical neck is the insertion site. Vidacare recommends the 45 mm needle on patients >40 kg. This is the preferred site for patients who are responsive to pain. Once the insertion is completed

secure the arm in place to prevent movement and accidental dislodgement of the IO catheter.

Identification of these insertion sites is critical for a successful insertion. Vidacare recommends regular review of the landmarks at each insertion site to assure a rapid and safe EZ-IO experience.

Clean the insertion site according to your institutional policy.

Slide 13



The insertion sites for pediatric patients are the same as those for adults. In small children, generally under the age of two, the tibial tuberosity may be difficult or impossible to locate. If the tibial tuberosity cannot be palpated, locate the distal aspect of the patella, move approximately 2 cm distal (depending on patient anatomy) and then medial to the flat aspect of the tibia.

At the distal tibia, locate the medial malleolus. Place one finger directly over the medial malleolus; move approximately 2 cm proximal (depending on patient anatomy) and palpate the anterior and posterior borders of the tibia to assure that your insertion site is on the flat center aspect of the bone.

Placement at the proximal humerus is the same as proximal humerus placement for adults. The site is located directly on the most prominent aspect of the greater tubercle. *Ensure that the patient's hand is resting on the abdomen and that the elbow is adducted (close to the body).* Slide thumb up the anterior shaft of the humerus until you feel the greater tubercle, this is the surgical neck. Approximately 1 cm above the surgical neck (depending on patient anatomy) is the insertion site. In small children, the proximal humerus may be difficult to locate. Therefore, the placement may be a proximal humeral shaft placement.

With all patients, minimal pressure should be placed on the driver during insertion. This is especially true with pediatric patients. With softer and smaller pediatric bones, special care must be taken during insertion to avoid both excessive pressure and recoil. Recoil can occur when the clinician feels the lack of resistance upon entry into the medullary space and inadvertently pulls back on the driver. This recoil may displace the needle set from the medullary space.

To avoid accidental dislodgement, Vidacare recommends stabilization of the EZ-IO once placement is confirmed.

Slide 14

When you are ready to place the needle, remove the EZ-IO driver from the case and remove the trigger guard if present.

Select the appropriate Needle Set and inspect the packaging and cartridge for damage. To assure sterility, verify that the safety seal is intact over the top of the needle cartridge.

Open the EZ-Connect tubing that comes with the Needle Set. If the patient is unresponsive to pain, attach a 10ml saline flush to the EZ-Connect and prime the tubing with saline, leaving the syringe and remaining saline attached to the EZ-Connect.
If the patient is responsive to pain, you need to administer 2% lidocaine without preservatives or epinephrine (cardiac lidocaine) for anesthesia prior to your initial saline flush. Prime the EZ-Connect with the appropriate amount of Lidocaine.

Once your EZ-Connect is primed, clean the insertion site according to your institutional policy for aseptic procedures.

Pop open the top of the cartridge to break the safety seal. Insert the driver into the cartridge to magnetically attach the driver to the Needle Set. Leave the cap on the needle until the site has been properly prepped and you are ready to complete the insertion.

Inspect the cartridge to ensure sterility is intact and that the cartridge was not opened prior to you:

-Intact sticker across the top.

-No red “2” is visible on top of cartridge.

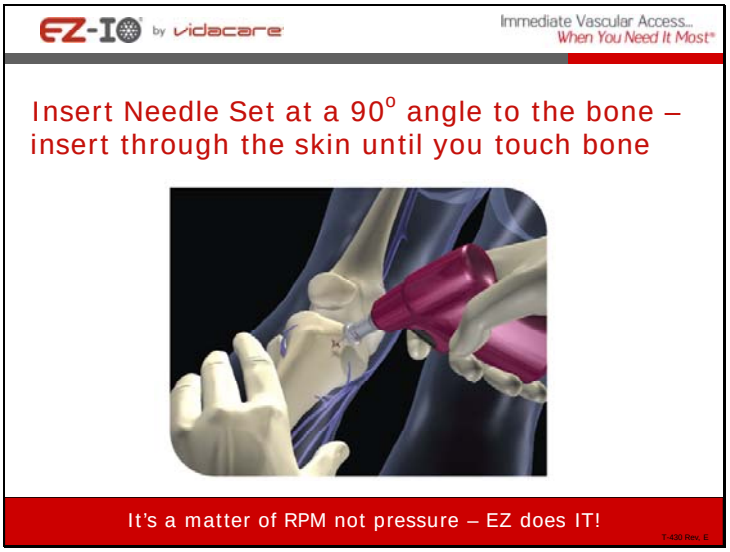
Slide 15

Once the site has been cleaned, and the EZ-Connect primed, remove the safety cap from the Needle Set by holding the needle cap firmly while momentarily powering the driver on. Allow the Needle Set to stop turning and then remove the Needle Set cap as previously mentioned.

Slide 16

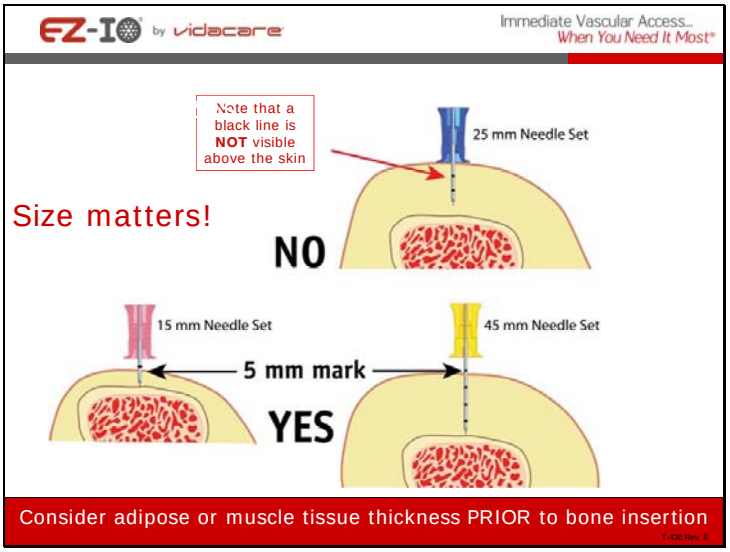
Stabilize the extremity to guard against unexpected patient movement.

Slide 17



Insert the needle set through the skin until you feel the tip of the needle touch the bone. This can be accomplished by powering through the skin and up to the bone or pushing through the skin and up to the bone. At this point you need to verify that you can see one of the black lines on the needle above the surface of the skin prior to powering the driver on. This assures that your Needle Set is long enough to penetrate completely through the compact bone and into the medullary space.

Slide 18

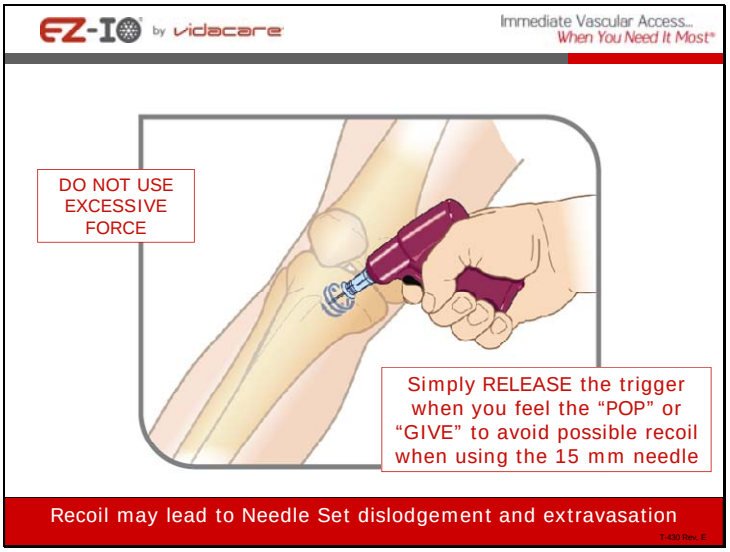


As this slide illustrates, patients have varying depths of tissue overlying the insertion site. This affects selection of the appropriate Needle Set. Though the amount of overlying tissue varies, adult compact bone is, on average, 3mm thick. In order to assure the Needle Set you have chosen is long enough; you must confirm visualization of a black line above the surface of the skin prior to powering the EZ-IO driver into the bone.

As you can see with the two illustrations on the bottom of the slide, you can visualize a black line above the surface of the skin with the needle tip resting against the bone. This check assures both of these Needle Sets are long enough to penetrate through the compact bone and reach the medullary space.

In the illustration at the top of the slide, the needle tip is not yet resting against the bone, and none of the black lines can be visualized. This is an indication that the selected Needle Set is not long enough for insertion at this site. Either a longer Needle Set must be selected for insertion at this site, or choose an alternative insertion site. Keep in mind the Needle Set can only be used once. If you are selecting an alternative insertion site, a new Needle Set must be used.

Slide 19



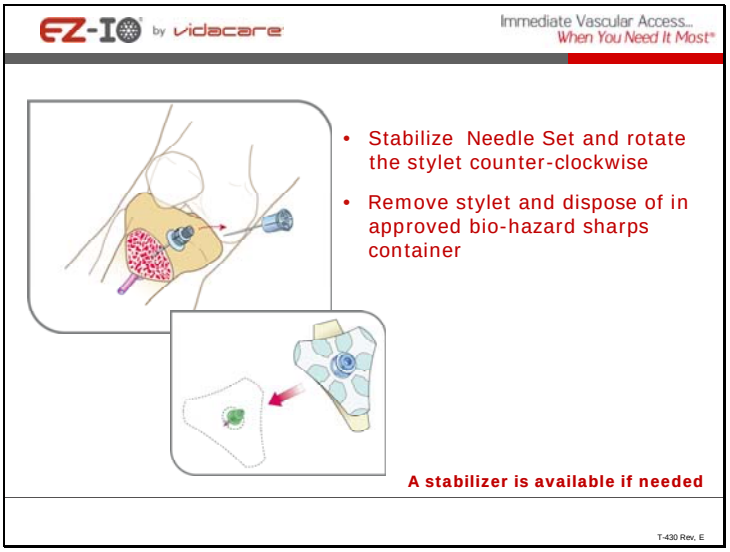
Once you have verified that the Needle Set is long enough for successful cannulation of the medullary space, you are ready to complete the insertion. While maintaining a 90° angle to the bone and keeping a light grip on the driver, power the driver on by depressing the trigger continuously. Apply the minimal amount of pressure required to keep the driver advancing straight into the bone. Remember the EZ-IO needle is advanced into the bone by utilizing RPMs rather than pressure. Maintain approximately 5 to 8 lbs. of pressure for an adult insertion to maximize the speed of insertion. With pediatric patients, minimal pressure is required for successful placement. Insertion time varies as bone density varies from patient to patient. Insertions in the tibia may take up to 10 seconds while placement in the proximal humerus may be slightly faster. In some patients insertion may take greater than 10 seconds, if the driver sounds like it is slowing down during insertion; reduce pressure on the driver to allow the RPMs of the needle tip to do the work.

In the unlikely event that the battery on the Power Driver fails clinicians may manually finish inserting the EZ-IO Needle Set. The Needle Sets were specifically designed to allow for this and the procedure merely requires the clinician to grasp the Needle Set and, rotate their arm, while pushing the needle into the intraosseous space. This may take several minutes.

Slide 20

When you feel a decrease in resistance indicating the Needle Set has entered the medullary space, take your finger off the trigger, and remove the driver by stabilizing the hub with one hand and pulling straight back with the driver. This releases the magnetic connection between the driver and the Needle Set. Be careful to avoid excessive movement of the Needle Set. The Needle Set should feel secure and stable in the bone with the driver removed.

Slide 21



While continuing to hold the needle hub, twist the stylet out of the needle by rotating the stylet counter-clockwise. Once removed, dispose of the stylet in an approved bio-hazard sharps container.

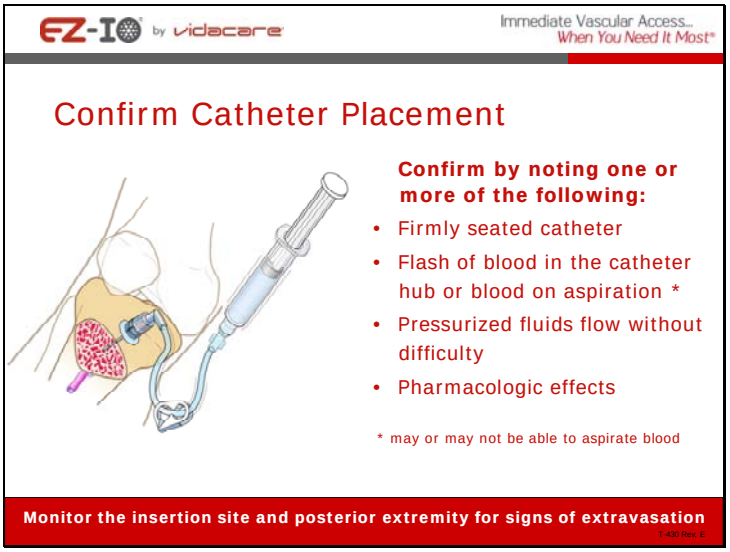
If using an EZ-Stabilizer®, apply the stabilizer prior to attaching the primed EZ-Connect tubing.

Once the stabilizer is in place, attach the EZ-Connect tubing to the needle hub.

Slide 22

Do not use the cartridge as a sharps container. The EZ-IO stylet will fit into most approved bio-hazard sharps containers. Using the cartridge as a sharps container may result in the inadvertent insertion of a contaminated stylet into a subsequent patient.

Slide 23

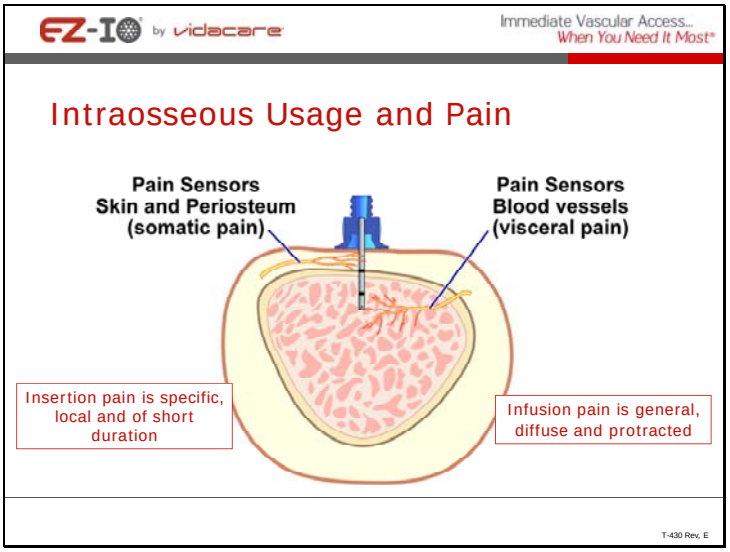


Confirmation of catheter placement can be achieved by aspirating marrow prior to the initial saline flush or lidocaine administration. Slowly retract the plunger on the syringe to withdraw marrow. If marrow is present, the needle has been successfully placed in the medullary space. Blood may also be noted in the hub of the needle after removal of the stylet. Absence of blood or inability to withdraw aspirate at the catheter hub does not mean the insertion was unsuccessful.

A firmly seated catheter and the ability to administer pressurized fluids without difficulty are indicators of successful cannulation of the medullary space; as is noting the pharmacologic effects of your medication administration.

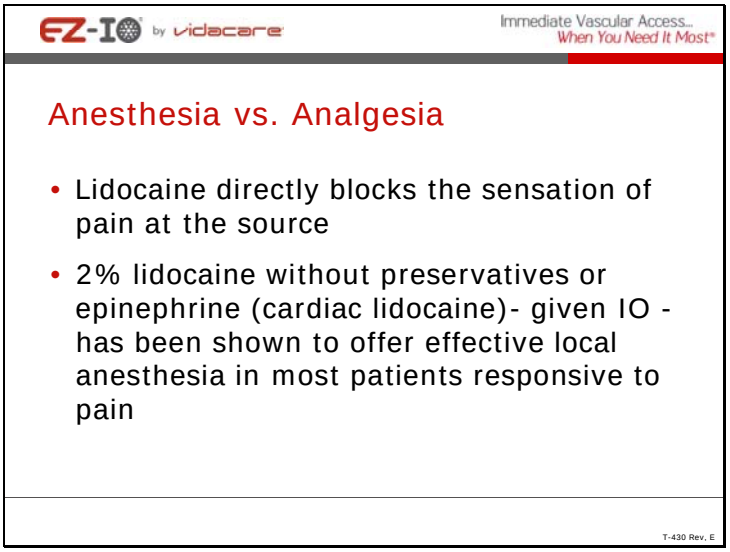
Once catheter placement has been confirmed, the site should be continually re-evaluated for signs of extravasation, fluid leakage or any other signs that indicate the needle tip is no longer in the medullary space.

Slide 24



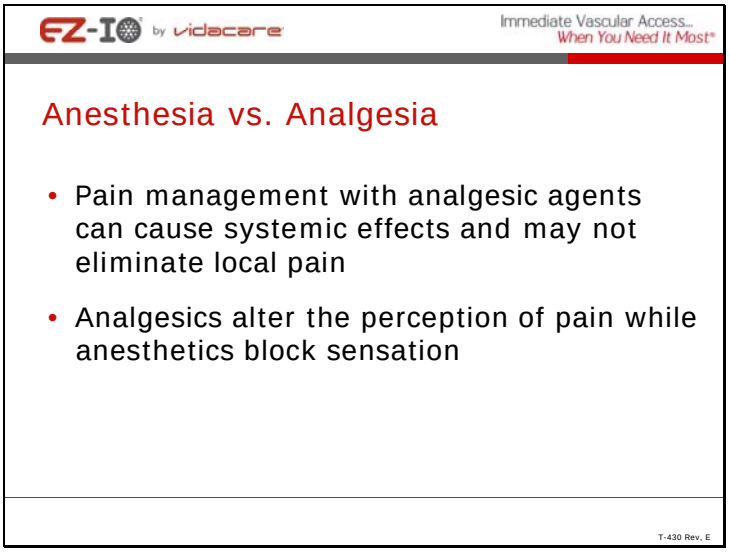
It is a common misconception that the insertion of the IO is the primary cause of pain. Anecdotally, piercing of the skin and actual insertion into the medullary space is reported as 2-3 on a 10 point pain scale. Many patients and volunteers rate the pain of insertion similar to, or less than that of insertion of an 18g peripheral IV. For patients able to perceive pain, there can be significant discomfort associated with the initial flush as well as infusion. Although the bone is a non-collapsible cavity it is also non-expandable. Therefore, rapid expansion and pressure changes occur within the bone as a result of fluid delivery.

Slide 25



Lidocaine is a well studied and common local anesthetic. Historically it has also been used as an anti-arrhythmic. Lidocaine is currently the drug of choice to numb or anesthetize the medullary space within the bone. Lidocaine in this use as a local anesthetic acts by inhibiting pain signals from reaching the brain. Lidocaine as an anesthetic is superior to analgesia for pain caused by stimulation of the pressure receptors inside the medullary space as it directly blocks the sensation of pain at the source. Lidocaine has the added benefit in that it will not cause respiratory depression or cause hemodynamic instability.

Slide 26



Pain management through analgesia such as narcotics does not block pain signals but rather it mitigates the sensations of pain throughout the body.

Slide 27

We have learned from volunteer studies that it is the absolute dosage, not the dosage per kg that counts. Because lidocaine is used for local anesthesia, not a systemic effect, it doesn't matter if you weigh 40 kg or 300 kg; the local vascular space in the bone we are attempting to anesthetize is about the same size.

If the patient responds to pain; 2% lidocaine without preservatives or epinephrine (cardiac lidocaine) has been demonstrated to be an effective local anesthetic for numbing the intraosseous space. As with any drug, always consult the pharmaceuticals DFU's prior to use and ensure the patient does not have an allergy or sensitivity to lidocaine.

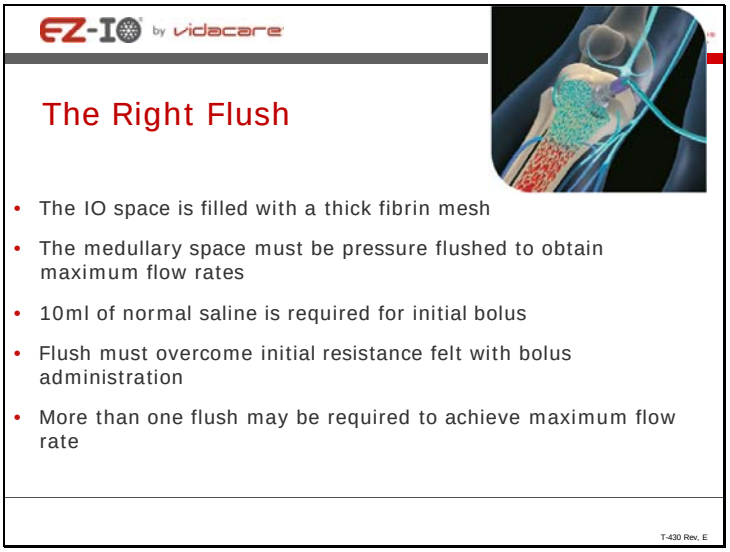
The prescribed dosage of 2% lidocaine without preservatives or epinephrine (cardiac lidocaine) must be infused slowly in 0.2 ml increments to prevent it from being sent directly into the central circulation. Titrated doses of the lidocaine should be given with increasing pressure as this will allow for expanded anesthetic effect in the medullary space. It is crucial that you wait a minimum of 15-30 seconds for the drug to reach maximum anesthetic effect before giving the bolus. Lidocaine is to be used as an anesthetic and not as analgesia and repeated dosing may be necessary.

Note: Medical Director must authorize appropriate dosage range (see Pain Management Bibliography M-220 for support). *The amount*

of lidocaine required to achieve pain relief in awake and responsive patients may vary based on individual differences and distracting injuries or conditions. Clinical correlation and judgment are required. Vidacare publication M-220 is an annotated bibliography of clinical research studies which address management of IO infusion pain and can be used as a resource for medical directors to determine appropriate the dosage range.

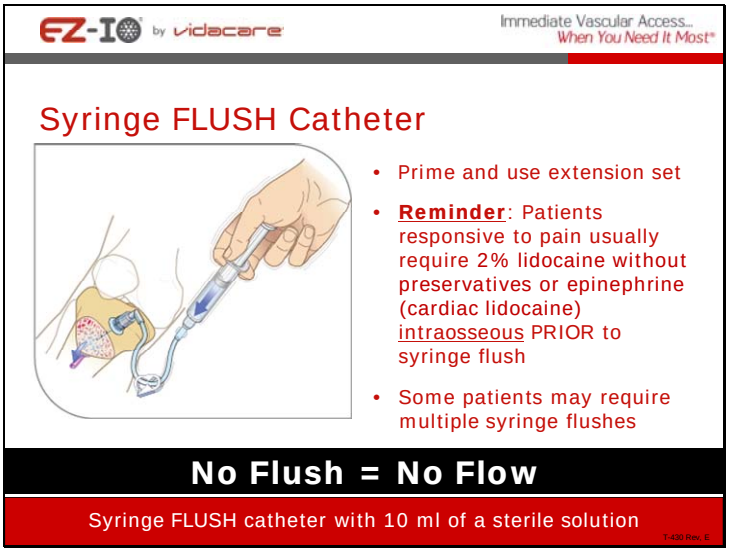
Do not exceed 3mg/kg/24hr

Slide 28



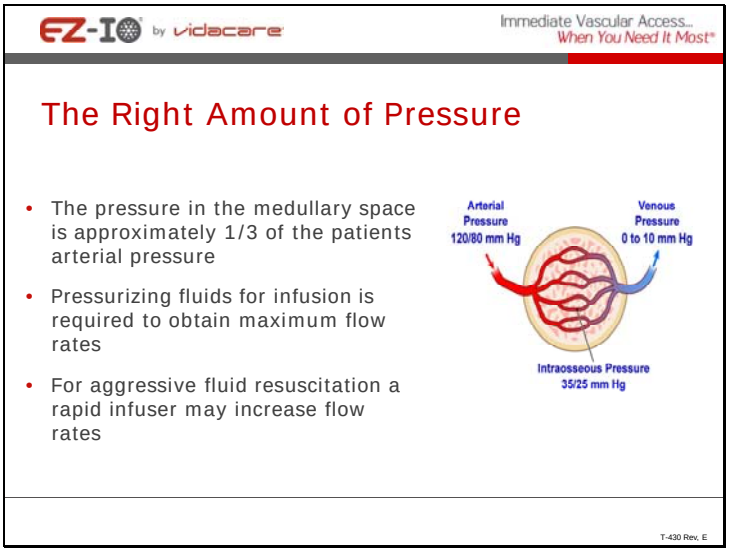
The intraosseous space is occupied with bone marrow which is held in place by a thick fibrin network. In order to obtain maximum flow rates you must displace this thick fibrin mesh. This is achieved with a rapid and vigorous 10ml flush with normal saline. The initial flush will be met with inherent resistance as the fibrin mesh is being displaced. This has been described as a similar sensation to pushing D50 or accessing a PICC line or vascular access port that has not been recently accessed. After the first vigorous flush of normal saline is given, IO flow rates are easy and rapid. Occasionally, patients require more than one flush or if IO has been saline locked for a prolonged period another vigorous flush may be required to obtain maximum flow rates.

Slide 29



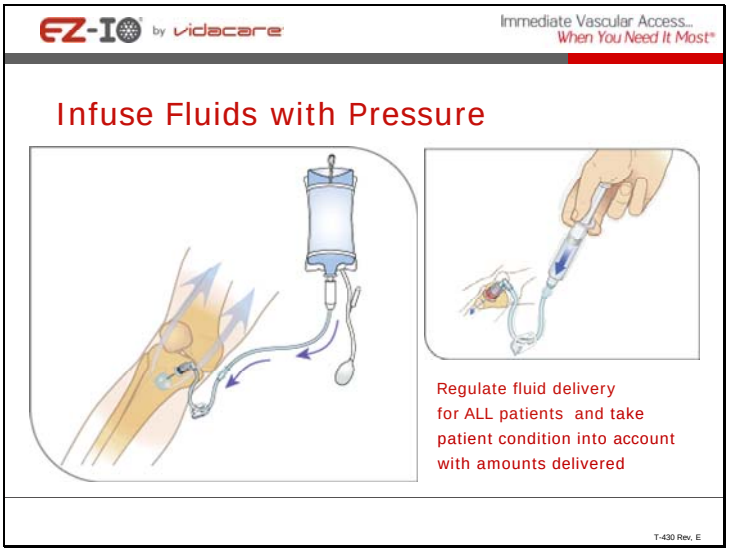
The EZ-Connect extension set should be primed with saline or 2% lidocaine without preservatives or epinephrine (cardiac lidocaine). If labs are to be obtained the tubing is not primed as blood will be withdrawn through the tubing. As a reminder, if the patient is responsive to pain the clinician should consider 2% lidocaine without preservatives or epinephrine (cardiac lidocaine) for anesthetic effect prior to the 10ml normal saline flush and it may be necessary to administer additional lidocaine following the saline flush. Remember: NO FLUSH, NO FLOW. If this step is omitted optimal flow rates will not be achieved.

Slide 30



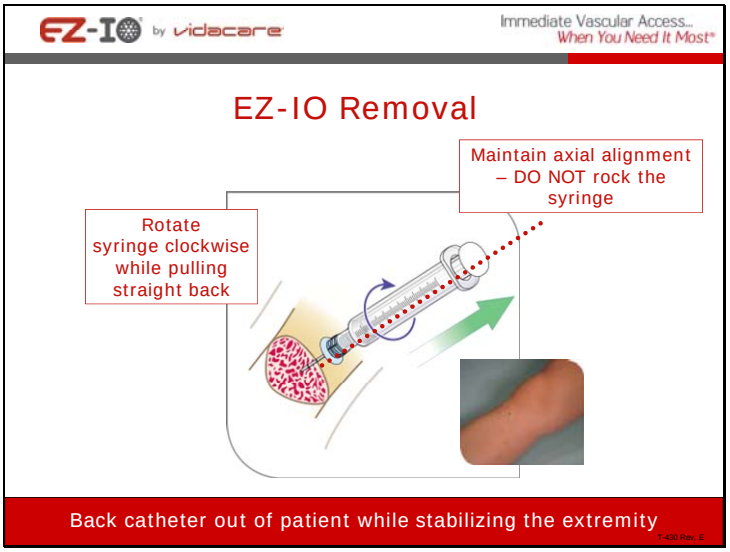
The pressure in the medullary space is approximately 1/3 of the patient's mean arterial pressure. This is important to remember because the pressure outside the bone in the IV bag must be higher than the pressure inside the bone to achieve flow. Therefore, fluids or medications must be delivered under pressure to obtain maximum flow rates. Rapid fluid infusers can be used for aggressive fluid resuscitation.

Slide 31



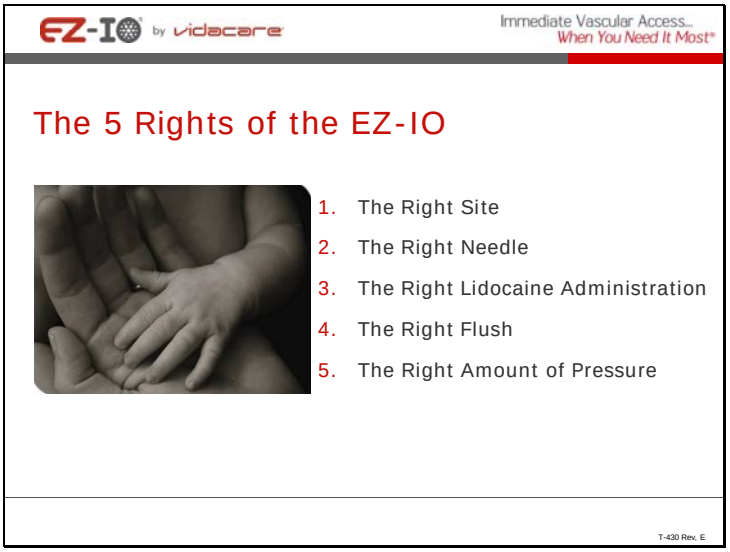
Although it is recommended that IO fluids be delivered under pressure it is not necessary that the fluids be delivered with an infusion pump or pressure bag. Fluids can also be delivered under pressure with a syringe. Patient condition should be taken into account when considering the amount of volume to be delivered.

Slide 32



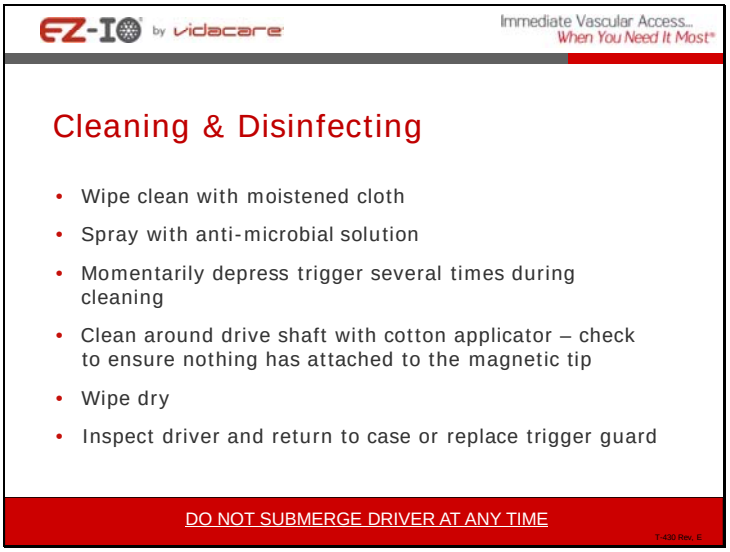
To remove the EZ IO begin by clamping the EZ-Connect tubing. The tubing is removed with counter- clockwise rotations of the Luer lock. Any size syringe with a standard Luer lock tip can then be attached to the EZ IO needle hub and will act as a handle for removing the IO needle. Stabilize the patient's extremity. The syringe that is attached to the IO hub is then rotated in a clockwise direction. This action releases the tight seal in the cortex. Continue rotating clockwise and pull straight out. It may take several rotations to remove the IO needle. Avoid rocking the needle. Bleeding at the IO site is anticipated to be minimal unless certain medications or medical conditions lead to increased bleeding. Local pressure at the site may be warranted in these circumstances. The site is cleaned with an approved skin antiseptic and in most cases only a simple adhesive strip or minimal dressing is required.

Slide 33



In review, all patients are entitled to the right size needle, the right insertion site, the right amount of 2% lidocaine without preservatives or epinephrine (cardiac lidocaine) if they are able to perceive pain, the right amount of flush and the right amount of pressure to obtain desired flow rates.

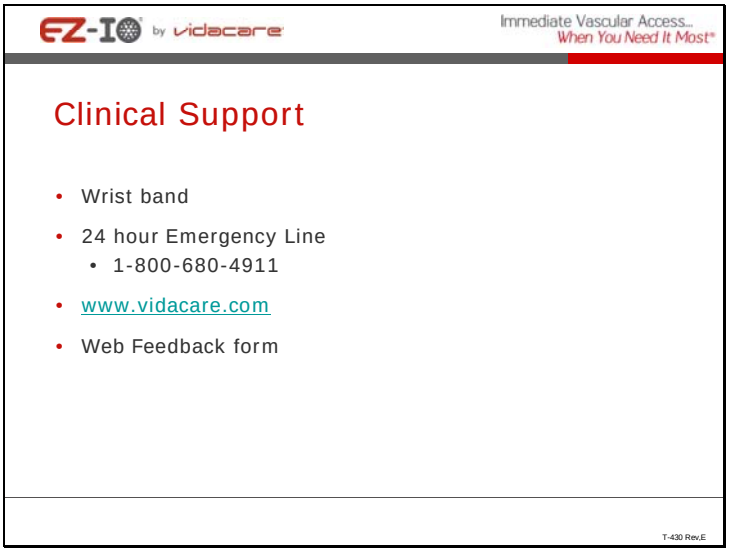
Slide 34



The EZ IO driver is manufactured using a non-porous fluid resistant medical-grade plastic; however, it is not fluid proof. For cleaning purposes, the driver should never be submerged in a cleaning solution. The driver should be wiped clean with a moistened cloth or sprayed with an anti-microbial solution. Clean around the drive shaft with a cotton applicator and check to ensure that nothing has attached to the magnetic tip. The driver is then wiped dry, inspected and returned to the carrying case. The trigger guard may also be replaced if no carrying case is utilized.

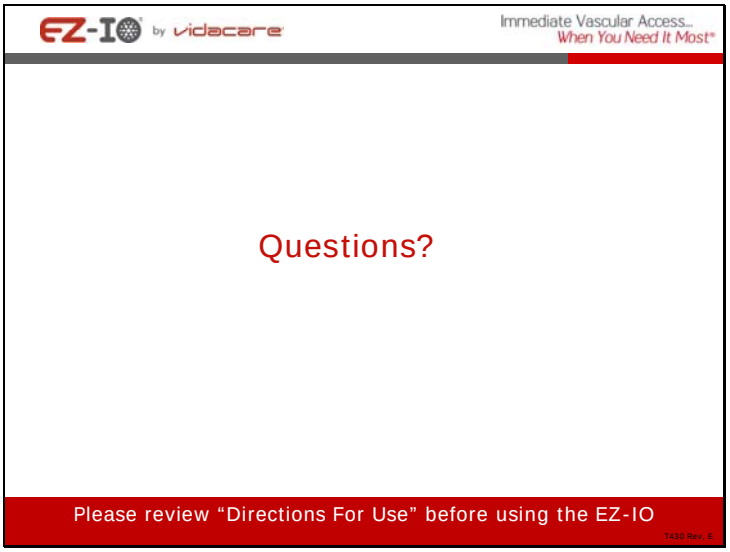
Refer to the Driver's DFU's for a more detailed instruction.

Slide 35



Vidacare provides 24/7 live clinical support to our clinical partners utilizing our 800 hotline (800-680-4911) which is featured on our patient wristbands, in addition to our website (www.vidacare.com). Vidacare places tremendous value on direct clinical feedback, so we invite you to visit our website to gain the newest, cutting edge product information as well as provide us with your direct feedback on our products performance.

Slide 36



If you have any questions please contact your Clinical Manager, you can find their contact information at www.vidacare.com or by calling 1-866- 479-8500. You may also contact the Director of Clinical Applications, Yvonne Bettis at yvonne.bettis@vidacare.com.